



HK IMO Training 2023–2024
Problem Set 15
For 27 January 2024



Problem 57. Find all polynomials $P(x)$ with real coefficients satisfying

$$4P(x)^2 + 4P(x) = 4P(x^2) + 1.$$

Problem 58. Let $n \geq 3$ be an integer. In how many ways can we put down 0 or 1 in each cell of an $n \times n$ table such that the sum of all six numbers in every 2×3 or 3×2 rectangle is even?

Problem 59. Let k be a positive integer. Prove that there exists a positive integer n such that the decimal representation of each of the numbers $n, n^2, \dots, n^k$ contains the consecutive digits 2024.

Problem 60. In $\triangle ABC$, let O , I and E be the circumcentre, the incentre and the A -excentre respectively. Let D be the intersection point of the line BC and the external angle bisector of $\angle BAC$. Let F be a point on the extension of IA beyond A such that $AF = 2IA$. Define P to be the point on the circumcircle of $\triangle DEF$ such that DP is a diameter. Prove that $OP = 3OI$.